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Semester: III	**Year : AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No : 19	
TITLE : JDBC SELECT Operation	

Problem Statement

Develop a Java application to establish connectivity with a database and perform data retrieval using the SELECT operation.

Learning Objectives

To establish database connectivity and retrieve records using JDBC SELECT statements.

Exercise

1. Create a Java application to display student records from database using SELECT query

Code:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

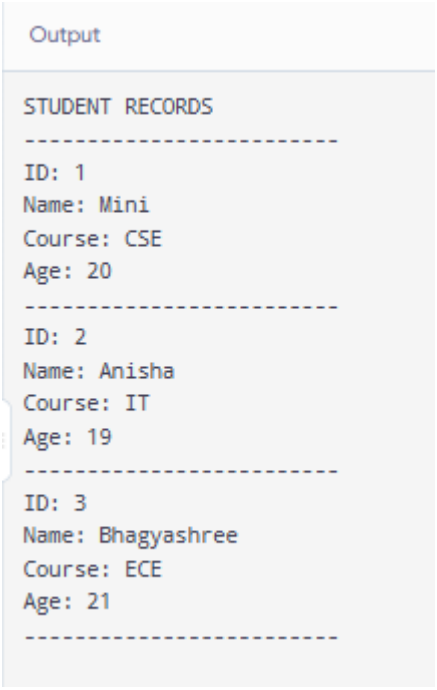
        String[][] student = {
            {"1", "Mini", "CSE", "20"},
            {"2", "Anisha", "IT", "19"},
            {"3", "Bhagyashree", "ECE", "21"}
        };

        System.out.println("STUDENT RECORDS");
        System.out.println("------------------------");

        for (int i = 0; i < student.length; i++) {
            System.out.println("ID: " + student[i][0]);
            System.out.println("Name: " + student[i][1]);
            System.out.println("Course: " + student[i][2]);
            System.out.println("Age: " + student[i][3]);
            System.out.println("------------------------");
        }

        sc.close();
    }
}
```

Output:



2. Create a Java application that connects to a database and displays product detailsusing a SELECT query. Show product ID, product name, quantity, and price.

Code:

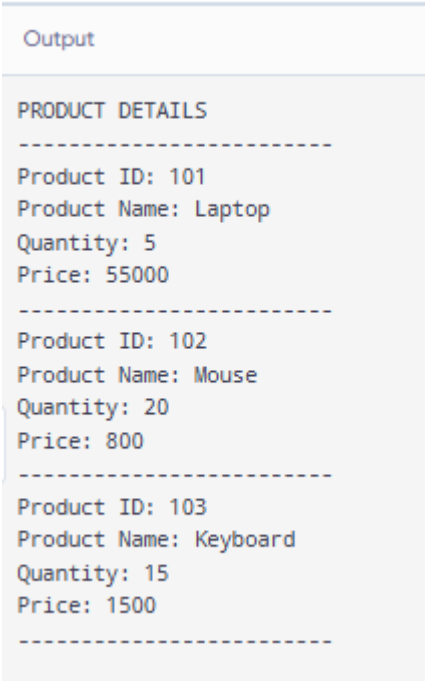
```
public class Main {
    public static void main(String[] args) {

        String[][] product = {
            {"101", "Laptop", "5", "55000"},
            {"102", "Mouse", "20", "800"},
            {"103", "Keyboard", "15", "1500"}
        };

        System.out.println("PRODUCT DETAILS");
        System.out.println("-------------------------");

        for (int i = 0; i < product.length; i++) {
            System.out.println("Product ID: " + product[i][0]);
            System.out.println("Product Name: " + product[i][1]);
            System.out.println("Quantity: " + product[i][2]);
            System.out.println("Price: " + product[i][3]);
            System.out.println("-------------------------");
        }
    }
}
```

Output:



Conclusion

Both programs demonstrate the concept of retrieving records from a database using Java and the SELECT query. The Student Records program displays details such as student ID, name, course, and age, while the Product Details program displays product ID, product name, quantity, and price. The JDBC concept is used to establish a connection between Java and the database and execute the SELECT query. The ResultSet is used to retrieve and display the records returned by the query. These programs help in understanding database connectivity, SQL SELECT queries, and retrieving data in Java. The programs also demonstrate how stored records can be accessed and displayed in a structured manner. Overall, they provide a basic understanding of connecting Java applications with databases and retrieving information from tables.